Class Lesson: Limits – From Average Speed to Instantaneous Speed

> ***Objective***
>
> *By the end of this lesson, you will understand:*
>
> - *What a limit is*
> - *Why limits were invented*
> - *How limits solve real-world problems*
> - *The connection between limits and average speed*
> - *How to solve limit problems*
> - *The foundation of derivatives*

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1. Motivation

Imagine you're driving a car.

Someone asks:

> ***"What was your speed exactly at 10:00 AM?"***

Not your average speed over the trip.

Not your average speed over the last hour.

Your speed **at that exact instant**.

Can we calculate it?

We only know how to calculate Average Speed

We know

$$\text{Average Speed} = \frac{\text{Distance}}{\text{Time}}$$

Example

You travel **120 km** in **2 hours**.

$$\text{Average Speed} = \frac{120}{2} = 60 \text{ km/h}$$

Easy.

But notice something important.

Average speed always uses a **time interval**.

It never tells us the speed **at one exact instant**.

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2. Why Average Speed Isn't Enough

Suppose you drove like this.

Time	Speed
9:00–9:30	40 km/h
9:30–10:00	80 km/h

Total distance

```
40 × 0.5 = 20 km

80 × 0.5 = 40 km

Total = 60 km
```

Average speed

$$\frac{60}{1} = 60 \text{ km/h}$$

Question:

> *Were you driving at **60 km/h** at 10:00?*

No.

You were driving at **80 km/h**.

Average speed hides what happened during the trip.

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3. Getting Closer to the Instant

Suppose we calculate the average speed over smaller and smaller time intervals before 10:00 AM.

Interval	Average Speed
10 min	60 km/h
5 min	61 km/h
1 min	62.5 km/h
30 sec	62.8 km/h
10 sec	62.95 km/h
1 sec	62.998 km/h

Notice something.

The numbers seem to settle toward

```
63 km/h
```

Even though we never measured the speed at exactly 10:00 AM.

Instead,

the average speed **approaches**

63 km/h.

This idea of **approaching** is exactly what a limit describes.

4. What is a Limit?

A limit answers the question:

> ***What value is something getting closer and closer to?***

Instead of asking

> *What is it right now?*

we ask

> *What does it approach?*

Mathematically,

$$\lim_{x \to a} f(x) = L$$

means

> *As **x gets closer to a**, the function gets closer to **L**.*

Notice

- x doesn't have to equal a
- we're only interested in what happens nearby

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5. Introducing Δ (Delta)

Suppose

At time

```
10:00
```

the car is at

```
200 km
```

One second later

```
200.0175 km
```

Distance traveled

$$\Delta s = 0.0175 \text{ km}$$

Time taken

$$\Delta t = 1 \text{ second}$$

Average speed becomes

$$\frac{\Delta s}{\Delta t}$$

This is the average speed over a very small interval.

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6. Shrinking the Time Interval

Now make the interval smaller.

Instead of

```
1 second
```

try

```
0.1 second
```

Then

```
0.01 second
```

Then

```
0.001 second
```

Every time,

compute

$$\frac{\Delta s}{\Delta t}$$

The answer gets closer and closer to one value.

That value is the **instantaneous speed**.

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7. The Birth of the Limit

We write

$$\boxed{\lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t}}$$

Read as

> *The limit of average speed as the time interval approaches zero.*

Notice something very important.

We never let

$$\Delta t = 0$$

We only let it get **closer and closer** to zero.

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8. Why Can't We Simply Use Zero?

Suppose we try.

Average speed

$$= \frac{\Delta s}{\Delta t}$$

At exactly one instant

$$\Delta s = 0$$

and

$$\Delta t = 0$$

giving

$$\frac{0}{0}$$

This is **undefined**.

So we never divide by zero.

Instead,

we ask

> *What happens as the denominator gets extremely close to zero?*

That's why limits exist.

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9. Visual Intuition

Imagine zooming in.

```
10 min interval


------------------------
Average = 60
```

Zoom in.

```
1 min interval

-----
Average = 62
```

Zoom again.

```
1 sec interval

-
Average = 62.99
```

Keep zooming.

Eventually,

the average speed practically becomes the instantaneous speed.

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10. The Mathematical Definition

Suppose

$$s(t)$$

represents position.

Average speed from

$$t$$

to

$$t+h$$

is

$$\frac{s(t+h)-s(t)}{h}$$

where

$$h$$

is the time interval.

Now make

$$h$$

smaller and smaller.

$$\boxed{\lim_{h \to 0} \frac{s(t+h) - s(t)}{h}}$$

This is the **instantaneous speed**.

Later,

you'll learn this is also called the **derivative**.

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11. Example Problem

A car's position is given by

$$s(t) = 5t^2$$

Find the instantaneous speed at

$$t = 2$$

Step 1

Use average speed.

$$\frac{s(2+h) - s(2)}{h}$$

Step 2

Compute

$$s(2+h) = 5(2+h)^2$$

Expand

$$= 5(4 + 4h + h^2)$$

$$= 20 + 20h + 5h^2$$

Step 3

Compute

$$s(2) = 20$$

Step 4

Substitute

$$\frac{20 + 20h + 5h^2 - 20}{h}$$

Simplify

$$= \frac{20h + 5h^2}{h}$$

Factor

$$= \frac{h(20 + 5h)}{h}$$

Cancel

$$= 20 + 5h$$

Now we have no division by zero.

Step 5

Take the limit.

$$\lim_{h \to 0} (20 + 5h)$$

Substitute

$$= 20$$

Answer

$$\boxed{20}$$

The car's instantaneous speed at

$$t = 2$$

is

```
20 units/second
```

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12. Why Did We Cancel h?

Notice

Initially

$$\frac{20h + 5h^2}{h}$$

We are **not** plugging in

$$h = 0$$

Instead,

we simplify while

$$h \neq 0$$

Only **after** simplifying do we let

$$h$$

approach zero.

This is the key idea behind limits.

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13. Another Example

Evaluate

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$$

Step 1

Substitute

$$= \frac{0}{0}$$

Cannot proceed.

Step 2

Factor

$$= \frac{(x-2)(x+2)}{x-2}$$

Step 3

Cancel

$$= x + 2$$

Step 4

Take the limit

$$2 + 2 = 4$$

Answer

$$\boxed{4}$$

14. Left-Hand and Right-Hand Limits

Suppose

```
Approach 3 from the left

2.9

2.99

2.999
```

This is written as

$$\lim_{x \to 3^-}$$

Approaching from the right

```
3.1

3.01

3.001
```

is written as

$$\lim_{x \to 3^+}$$

If both sides approach the same value,

the limit exists.

Otherwise,

it doesn't.

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15. Common Indeterminate Forms

When substitution gives

$$\frac{0}{0}$$

don't panic.

It simply means

> *"More work is required."*

Common techniques include

- Factoring
- Rationalization (using conjugates)
- Trigonometric identities
- L'Hôpital's Rule (later)

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16. Common Misconceptions

❌ A limit means plugging in the value.

No.

A limit asks what happens **near** the value.

❌ The function must exist there.

No.

A function may be undefined at a point while its limit still exists.

Example

$$\frac{x^2-4}{x-2}$$

is undefined at

$$x=2$$

yet

$$\lim_{x\to 2}\frac{x^2-4}{x-2}=4$$

❌ Approaching means eventually reaching.

Not necessarily.

A limit only cares about what values get arbitrarily close to, not whether they ever equal that value.

17. Summary

A **limit** is a mathematical tool for describing what a quantity approaches as its input gets closer to a specific value.

The concept arose naturally from the need to define **instantaneous speed**. Since average speed requires a time interval, we make that interval smaller and smaller. Although we never let the interval become exactly zero (to avoid division by zero), the average speed approaches a single value. That limiting value is the instantaneous speed.

In general:

- Average speed:

$$\frac{\Delta s}{\Delta t}$$

- Instantaneous speed:

$$\lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t}$$

- General limit notation:

$$\lim_{x \to a} f(x) = L$$

Limits are the foundation of **differential calculus**. They lead directly to derivatives, rates of change, optimization, differential equations, and many of the mathematical tools used in physics, engineering, economics, and computer science. Understanding limits is the first major step toward understanding calculus itself.